

MC-Core Block & Port Reference

One channel, AXI → DFI. Every sub-block: ports (dir/width), valid-ready, and [instance count]. Params: N_RANKS(=2), N_BANKS=32 (8 BG×4), N_BG=8, BL=16, GEAR ∈ {1, 2, 4}, ROB_IDX=8b, 64B-aligned addr.

1 Boundary interfaces

AXI write channels [×1]

Port	Dir	Width	Meaning / valid-ready
awid	in	AXI_ID_W	write addr id
awaddr	in	ADDR_W	64B-aligned (LSB[5:0]=0)
awlen/awsize/awburst	in	8/3/2	beats-1 / bytes-per-beat / burst type
awvalid/awready	i/o	1/1	AW handshake
wdata/wstrb	in	512/64	write beat + byte enables (DM)
wlast/wvalid/wready	i/o	1	W handshake; wlast = last beat
bid/bresp/bvalid/bready	o/o/o/i	ID/2/1/1	write response (self-timed)

AXI read channels [×1]

Port	Dir	Width	Meaning / valid-ready
arid/araddr/arlen/arsize/arbust	in	ID/ADDR/8/3/2	read request
arvalid/arready	i/o	1	AR handshake
rid/rdata/rresp/rlast	out	ID/512/2/1	read return beats
rvalid/rready	o/i	1	R handshake

DFI egress [×1] (to PHY)

Port	Dir	Width	Meaning / valid-ready
dfi_address	out	14×GEAR	CA bus, per phase (2-cyc cmds)
dfi_cs_n/dfi_act_n	out	(2/1)×GEAR	chip-select / activate per phase
dfi_bg/dfi_bank	out	(3/2)×GEAR	bank group / bank
dfi_wrdata/_mask/_en	out	(2·32/16/1)×GEAR	write data + DM + enable
dfi_rddata/_valid	in	(2·32/1)×GEAR	read data + valid (RL later)
dfi_cke/dfi_odt/dfi_reset_n	out	N_RANKS	power / ODT / reset
dfi_freq_ratio/_init_start	out	2/1	gear + freq-change handshake
dfi_init_complete/dfi_alert_n	in	1/1	PHY ready / write-CRC alert

2 Request admit + write ROB [×1]

req_validate + aw_id_alloc [×1]

Port	Dir	Width	Meaning / valid-ready
axi_req_in	in	-	{axi_id, ddrphy_addr, burst_len}, req valid
narrow	out	1	size<full wstrb≠1s sub-64B
req_out	out	-	{aw_id, axi_size, axi_len, w_phy_addr}, valid/ready

write_rob [×1] (depth ROB_DEPTH)

Port	Dir	Width	Meaning / valid-ready
alloc_in/alloc_we	in	-/1	{aw_id, addr, narrow}; admit iff !full
full	out	1	back-pressures AW
pkt_in_pulse	in	1	++gen_idx / packet landed
wr_done_pulse	in	1	++done_cnt (from write_rob_done)
N/gen_idx/done_cnt	reg	8×3	per aw_id: total / addr-index / retire
retire	out	1	(done_cnt==N) --> mc_resp, free

tcam {aw_id} [×1] (depth ROB_DEPTH)

Port	Dir	Width	Meaning / valid-ready
lookup_aw_id	in	AW_ID	CAM match
idx/hit	out	log2(DEPTH)/1	row index + valid

3 Ingress lane [×N_RANKS]

rd_bp_fifo / wr_bp_fifo [×N_RANKS each] (depth 16)

Port	Dir	Width	Meaning / valid-ready
push/pop	in	1	write/read enable
din/dout	i/o	-	{rob_index, phy_addr, (wr:wd_slot)}
full/empty/partial_full	out	1	flags --> turn_ctrl

turn_ctrl + wlr_sel [$\times N_RANKS$]

Port	Dir	Width	Meaning / valid-ready
rd/wr_full_flags	in	2	from bp-fifos
batch_dir	reg	1	R/W batch direction
batch_cnt	reg	W	batch/age counter (cap)
sel_out	out	1	rd/wr FIFO select (combined path)
combined_req/valid	out	-/1	one merged path --> lookahead

lookahead [$\times N_RANKS$] (4-entry)

Port	Dir	Width	Meaning / valid-ready
e0..e3	reg	4×-	shift-reg window (e0=head)
bank_rdy	in	32	from scoreboard/bank states
bank_cmp	int	6	kill younger same-bank
head_out/sel	out	-/1	prio-enc winner; shift on sel

bank_dec [$\times N_RANKS$]

Port	Dir	Width	Meaning / valid-ready
phy_addr	in	-	decode
bank_oh	out	32	1-of-32 {bg,bank}; drives cell WE

4 Per-bank + arb [$\times N_RANKS \times 32 = 64$]

bank_cell / FSM_TABLE [$\times 64$] (= pre_bank_elig repeated 32/rank)

Port	Dir	Width	Meaning / valid-ready
state	reg	4	IDLE/ACTING/OPEN/PREING/REFING/...
valid/open_row/pre_pending	reg	1/18/1	status
last_op/ref_pending	reg	1/1	dir / refresh flag
bank_req_reg	reg	~28	{rob_index,col,op,sram_addr}
can_act/cas/pre	in	3	from scoreboard (SR flops)
need_act/cas/pre,is_hit	out	4	row_comparator result
bank_go/bank_cmd	out	1/2	elig → cmd_arb
grant_oh	in	1	pop-on-CAS from L1/L2

cmd_arb [$\times 64$]

Port	Dir	Width	Meaning / valid-ready
need_*, can_*, is_hit	in	-	eligible candidates
cas_w/act_w/pre_w	reg	W	weights (row-lock / age / close)
bank_cmd/bank_go	out	2/1	W_MAX pick → bgb_arb

bgb_arb [$\times N_RANKS \times 8 = 16$]

Port	Dir	Width	Meaning / valid-ready
b0..b3_go/cmd	in	4×	4 banks in the BG
p_bit[4]	in	4	ref-drain priority (MSB)
bank_winner/vld	out	2/1	4→1 (weight+p_bit)

bg_arb [$\times N_RANKS = 2$]

Port	Dir	Width	Meaning / valid-ready
bg0..bg7_winner/vld	in	8×	8 BG winners
last_cas_bg	in	3	diff-BG rotation bias
rank_winner/vld	out	-/1	8→1 per rank

l3 class-split [×1]

Port	Dir	Width	Meaning / valid-ready
r0/r1 cas/act/pre	in	6×	per-rank class winners
last_cas_dir/rank	in	1/1	streak + rank-hop
best_cas/act/pre	out	3×	≤3 grants → packer

5 Scoreboard / timestamp

writeback__upd [×1] (adders replicated per cmd×rel)

Port	Dir	Width	Meaning / valid-ready
grants	in	≤3+ref	{rank,bg,bank,phase_off}
gc	in	13	CK timebase (+ = gear)
timing_const	in	~20×10	shared JEDEC addends
next_*/we_*	out	13/-	per-scope deadline + write-enable
reset_*	out	-	per-scope can_* SR reset (issue-decode)

scope reg-files (flop arrays, parallel-read)

Port	Dir	Width	Meaning / valid-ready
bank_treg	reg	64×39	next_act/cas/pre [×64]
bgs_treg	reg	16×26	next_act_bg/cas_bg [×16]
bgd_treg	reg	2×13	next_act_dbg [×2]
rank_treg	reg	2×85	faw_ts[4]/next_rd/wr/gates [×2]
global_reg	reg	54	dqFree/caFree/last_cas [×1]

comparator + SR flop [×(64·3 + 16·2 + 2 + ...)]

Port	Dir	Width	Meaning / valid-ready
ge	int	-	(gc-next)[MSB]==0 → SET
reset	in	1	issue-decode (reset-dominant)
can_x	out	1	1-bit flop → arb

6 Phaser [×1 each]

phase__packer [×1]

Port	Dir	Width	Meaning / valid-ready
grants/me_cmd	in	≤3/1	demand + maint flag
gc/gear	in	13/2	phase offsets, slot mask
tokens	out	≤4	{cmd,bank,addr,phase_off,len}
commit_pulse	out	≤4	FSM/scoreboard/RAA writeback

ca_beat__encoder [×1]

Port	Dir	Width	Meaning / valid-ready
tokens	in	≤4	from packer
tail_pending	reg	1	1:1 straddle carry
dfi_address/cs_n/act_n	out	per phase	2-cyc → 2 beats

dfi_mux [×1]

Port	Dir	Width	Meaning / valid-ready
init_drive/sched_drive	in	2×	two DFI sources
init_done	in	1	sel (one-way latch)
dfi_out	out	-	to PHY

7 Data path [×1 each]

wr__accum [×1]

Port	Dir	Width	Meaning / valid-ready
w_beats/wstrb/narrow	in	512/64/1	AXI beats + strobe + flag
line_512/strobe	out	512/64	64B packet + DM mask

Port	Dir	Width	Meaning / valid-ready
line_done	out	1	wlast / 64B boundary

wd_sram [×1] / rd_sram [×1]

Port	Dir	Width	Meaning / valid-ready
wr_ptr/rd_ptr	in	PTR	alloc-on-arrival / RL-line
line_io	i/o	512	one packet per line
full	out	1	back-pressure W

wr_split+wl_launch / rd_cap+rd_accum [×1 each]

Port	Dir	Width	Meaning / valid-ready
arm_in	in	-	{op,rob_index,sram_addr,beats} @CAS-commit
dfi_wrdata / dfi_rddata	o/i	2·32×GEAR	beats (CWL / RL delayed)
done	out	1	wr_done (self-timed) / rd_done (last beat)

completion [×1]

Port	Dir	Width	Meaning / valid-ready
rd_done/wr_done	in	2	two done sources
resp_afifo_out	out	13	{rob_index, pkt_num, status} → CIF

8 Maintenance engine [×1] (refresh timers ×N_RANKS)

Seven sub-FSMs (Init/Refresh/RFM/ZQcal/MR_Poll/MR_Write/PwrMgmt) + MR_Read_Arb + FreqChange (stub) + Error (stub). Full port lists in fig_39-46. Common bus:

Port	Dir	Width	Meaning / valid-ready
me_cmd_valid/type/rank/bg/bank/mrout		1/3/./...	flag to scheduler
sched_ack	in	1	packer accepted
set/clr_gate_rfc/zq/mr	out	N_RANKS	scoreboard gates
raa_inc_en/raa_dec	i/o	16×N_RANKS	RFM counters
mrr_data_valid/data/rank/requester	in	1/8/./...	RD_CAP sideband
init_done	out	1	DFI mux handoff